

Review Protocol and Extraction Codebook

LLM-Based Test Oracles: Source-of-Authority Taxonomy - A Systematic Literature Review

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Criteria locked 2026-06-07; codebook fixed in advance

Inclusion / Exclusion Criteria (LOCKED)

Locked: 2026-06-07, before screening began.

Review: PRISMA-2020 SLR - LLM-based test oracles, organized by source of oracle authority.

Applied to: the 2,245 de-duplicated records (Screening-Funnel sheet, Review-Data.xlsx).

Inclusion - a record is included only if ALL hold

- **I1 - Oracle focus.** The work concerns a **test oracle** - the mechanism that decides expected behavior / correctness (assertions, expected outputs, properties/invariants, metamorphic relations, differential comparison, human judgment). Not merely test *input* generation or coverage.
- **I2 - LLM involvement.** A **large language model generates, infers, or determines the oracle**, OR the study **empirically examines how the oracle is determined / behaves** in an LLM-based testing pipeline.
- **I3 - Window.** Published Jan 2021 - May 2026.
- **I4 - Form.** ≥ 4 pages, English, full text retrievable, peer-reviewed or indexed (conference/journal/early-access).

Exclusion - a record is excluded if ANY holds

- **E1 - Testing of LLMs.** The LLM is the system under test (e.g., hallucination/bias/QA metamorphic testing of LLMs), rather than an LLM used *for* testing software.
- **E2 - No oracle aspect.** LLM test/code generation that does not foreground an oracle: test-input-only, coverage-only, prefix-only, program-repair-only.
- **E3 - Hardware verification.** Assertion/oracle generation for hardware (SystemVerilog/RTL/VLSI). Out of software scope. (*Noted as an adjacent line in future work.*)
- **E4 - Secondary study.** Surveys, SLRs, systematic mapping studies. Excluded as *primary* studies; tagged **secondary/survey** and retained in a separate "related reviews" list for the differentiation/related-work section (NOT counted in the included primaries, to avoid double-counting).
- **E5 - Non-peer-reviewed preprint.** arXiv-only / unpublished preprints. (*The search covered Scopus/IEEE/ACM only; preprint non-coverage is disclosed as a threat to validity.*)
- **E6 - Non-research / form.** Editorials, keynotes, posters, tool demos, tutorials, extended abstracts, < 4 pages, non-English, non-retrievable.

Operating notes (how the criteria are applied)

- **D5 rule (oracle vs input).** A test = input + oracle; this review is about the *oracle*. Papers that produce assertions / expected outputs / checked properties are IN; papers that only produce test inputs are OUT.
- **Page threshold (I4 / E6).** Applied pragmatically - page counts are not cleanly recorded for

every record (journals use article numbers); used mainly to exclude obvious posters / extended abstracts. Journal articles auto-pass.

- **Compositional systems.** A study may use multiple oracle sources; it is included if it meets I1-I4, and its primary + secondary source categories are recorded at extraction.
- **Screening fields.** Each record in the Screening-Funnel sheet gets a **decision** (include / exclude / maybe) and a **reason** (mapped to the criterion code above, e.g., E1, E2-no-oracle).

Provenance (decisions, 2026-06-07)

- D1 hardware -> **exclude** (E3). D2 surveys -> **exclude as primary, keep for related-work** (E4). D3 preprints -> **non-arXiv only** (E5). D4 page threshold -> **>=4pp** (I4). D5 oracle-vs-input rule -> **adopted**.

Extraction Codebook (data dictionary) - LLM-based test oracles SLR

Use: one full-text pass per included paper produces (i) an eligibility verdict and (ii) the coded fields below. Output = one row per paper id in the coding sheet. Controlled vocabularies are fixed in advance. Code **primary + secondary** where a paper is compositional; use **other:<free text>** when nothing in the vocabulary fits. In this public release the free-text evidence, quote, and notes fields are omitted; the coded categorical fields and citation metadata are retained.

A. Identity (pre-filled from worklist / dedup_master)

id / title / authors / year / venue / venue_type {conference, journal, workshop, preprint} / doi

B. Eligibility - Stage 4 (full-text)

- **ft_decision** {include, exclude}
- **ft_reason** (if exclude) {E1 testing-of-LLMs, E2 no-oracle, E3 hardware, E4 secondary, E5 preprint-only, E6 non-research/<4pp, I1-fail, I2-fail, I3-fail, I4-fail}
- **ft_confidence** {high, medium, low}
- **ft_evidence** - short quote + section locator

C. RQ1 - Landscape & tooling

- **domain** {general-SW, web, mobile, API/REST, GUI, ADS/autonomous, smart-contract, security, ML/DL-system, embedded/PLC, scientific/numeric, games, other}
- **sut_type** {unit/code, system/E2E, GUI, API, model, contract, other}
- **language** {Java, Python, C/C++, C#, JS/TS, Solidity, multiple, NA, other}
- **llm** {GPT-4/4o, GPT-3.5, Claude, Gemini, Llama/CodeLlama, CodeT5/CodeBERT, StarCoder, DeepSeek, multiple, other}
- **llm_access** {open, closed, both}
- **adaptation** (multi) {zero-shot, few-shot, CoT/reasoning, RAG, fine-tuning, pre-training, agent/multi-agent}
- **research_type** (Wieringa) {proposal, validation, evaluation, experience, philosophical}
- **artifact** {public, on-request, none}

D. RQ2 - Source of authority (core)

Where the verdict's correctness ultimately comes from.

- `source_primary` {implementation-derived, specification-derived, reference-differential, regression-prior-version, human-elicited, model-parametric, implicit-intrinsic}
- `source_secondary` {same vocab, or none}
- `spec_formality` (only if specification-derived) {formal, semi-formal, natural-language, NA}

E. RQ3 - Specification (in)dependence

- `spec_required` {yes, optional, no}
- `verdict_basis` {intended-behavior(spec), actual-behavior(implementation), mixed, NA} - the actual-vs-expected axis
- `spec_independence` {independent, partly, dependent} - derived judgement (independent = verdict needs no external spec)

F. RQ4 - Form

- `oracle_form` (multi) {assertion, metamorphic-relation, property/PBT, invariant, expected-output-value, classifier/judge-label, model-checked-property, other}

G. RQ5 - Adjudication mechanism & LLM role (core)

How the pass/fail is computed (kept separate from D).

- `mechanism` (multi) {exact/structural-match, semantic/similarity, LLM-as-judge, statistical/distributional, formal-verification, runtime-assertion-execution, human-review, other}
- `llm_role` {generates-oracle-artifact, is-the-oracle/judge, assists/refines, validates, NA}
- `pipeline_stage` {oracle-generation, test+oracle-generation, runtime-checking, oracle-repair, evaluation-only}

H. RQ6 - Evaluation & oracle quality

- `benchmark` (multi) {Defects4J, HumanEval, MBPP, SF110, Methods2Test, GitBug-Java, custom, none, other}
- `baseline` {yes(+name), no}
- `metric` (multi) {fault-detection, mutation-score, accuracy-vs-reference, precision/recall/F1, coverage, exact-match, BLEU/CodeBLEU, other}
- `ground_truth` {human, reference-impl, existing-tests, specification, none}
- `oracle_quality_method` - *how the oracle's OWN correctness is checked* {manual-inspection, vs-ground-truth-oracles, mutation, verification-in-loop, human-in-loop, not-assessed}

I. RQ7 - Trust & failure modes

- `failure_modes` (multi) {hallucination, weak/missing-assertion, over-assertion, flakiness/non-determinism, data-contamination, cost/latency, prompt-sensitivity, other}
- `mitigations` - short free text

J. RQ8 - Gaps (per-paper inputs to synthesis)

- `limitations` - short free text (author-stated)

- `future_work` - short free text (author-stated)

K. Free

- `notes` - anything notable
 - `key_quote` - 1 verbatim quote + locator that anchors the source/form/mechanism coding
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Coding rules

1. **Source mechanism.** D (authority) and G (mechanism) are independent - e.g., an LLM-as-judge given a spec -> `source_primary=specification-derived, mechanism=LLM-as-judge`.
2. **Compositional** studies: record `source_secondary`; multi-valued fields take a list.
3. **Confidence + evidence** required on the eligibility verdict and on `source_primary` (the contested scientific judgement that gets the human check).
4. **"other:" escape** used when nothing in the controlled vocabulary fits.